```
        registeredStudents[msg.sender].personNumber = personNumber;
   }

    function loginStudent(uint personNumber) public view
                                                returns (bool){        ❺
        if(registeredStudents[msg.sender].personNumber == personNumber){
            return true;
        }else{
            return false;
        }
    }

    function addPreRequisiteCourse(uint courseNumber, uint grade)
                                          public validStudent {        ❻


        if(courseNumber == 115) {
            registeredStudents[msg.sender].prereq115 = grade;
        }
        else if(courseNumber == 116) {
            registeredStudents[msg.sender].prereq116 = grade;
        }
        else {
            revert("Invalid course information provided");
        }

        ...    }

    function addCoreCourse(uint courseNumber, uint grade) public
                                                validStudent {         ❻
   { ...}

    function addDomainSpecificCourse(uint courseNumber, uint grade) public
                                         validStudent {                ❻

        ...}

    function addCapstoneCourse(uint courseNumber, uint grade) public
                                         validStudent {                ❻

        ...  }

    function checkEligibility(uint personNumber) public validStudent
                                          returns(bool) {              ❼


        ...
// courses in each category are examined and event emitted if satisfied
// overall GPA computed if all course requirements are satisfied
  if(registeredStudents[msg.sender].prereq115 > 0 &&
        registeredStudents[msg.sender].prereq116 > 0) {

            preRequisitesSatisfied = true;
            emit preRequisiteSatisfied(personNumber);
            totalGPA += registeredStudents[msg.sender].prereq115 +
                        registeredStudents[msg.sender].prereq116;
...
```